

Threats to Identification

An Honest Assessment of Our Causal Claim — Zehan Qin | ECON 5200 | Spring 2026

This memo honestly assesses what could go wrong with the causal claim made in the accompanying Technical Report — that the NSW job training program increases annual earnings by \$1,541 per participant. Each threat is accompanied by a bias direction analysis, proposed mitigation, and discussion of what we cannot rule out. The Double Machine Learning (DML) approach used here is defensible, but observational causal claims demand the critical honesty we provide below.

1. Most Serious Threat: Unobserved Confounding

The threat. DML assumes conditional independence: that, conditional on observed covariates (age, education, race, marital status, pre-treatment earnings), treatment assignment is independent of potential outcomes. However, NSW participants were referred through specific channels — probation officers, social workers, drug rehabilitation programs — based on characteristics we cannot observe in the data: motivation, mental health, criminal history, social networks, and willingness to comply with program rules. If these unobserved factors also affect future earnings, our conditional independence assumption fails and the DML estimate is biased.

Direction of bias. Ambiguous, but likely *upward* after DML correction. Participants selected for having the most barriers may have lower potential outcomes even after controlling for observables, which would make the treatment appear more effective than it truly is. Alternatively, if the most motivated disadvantaged workers self-selected into the program, the bias could flip direction. The close alignment between our RF-DML estimate (\$1,541) and the experimental benchmark (\$1,794) suggests unobserved confounding is not catastrophic in this dataset, though it cannot be fully ruled out.

What would address it. An instrumental variable that affects program participation but not earnings directly — difficult to find in this context. Alternatively, the original NSW RCT design (randomization within the eligible population) eliminates this threat entirely. The experimental benchmark of \$1,794 serves as a validity check that observational methods rarely have access to.

2. Second Threat: Overlap / Positivity Violation

The threat. The treated group (NSW participants) and control group (CPS workers) are drawn from very different populations. Many CPS workers have covariate values — high earnings over \$20,000, older age, higher education — that no NSW participant shares. The standardized mean differences confirm extreme imbalance: $|SMD| = 2.43$ for race, -1.57 for 1974 earnings, -1.23 for marital status. This violates the overlap/positivity assumption: we cannot reliably estimate treatment effects in regions of covariate space without common support.

Why it matters. DML may extrapolate into regions with no treated observations, producing unreliable estimates. The wide confidence intervals in our primary specification ($[\$229, \$2,853]$ with RF nuisance) partly reflect this extrapolation uncertainty.

Partial mitigation. We could trim the sample to the common support region — dropping CPS observations with propensity scores near 0 or above 0.99. This would reduce sample size but improve internal validity. A formal propensity score distribution check is a natural extension we

have not yet performed but would recommend for a production analysis.

3. Adversarial Angles

Specification Sensitivity

Our two DML specifications give meaningfully different point estimates: \$365 with Gradient Boosting nuisance models versus \$1,541 with Random Forest. This roughly fourfold difference reflects how ML model choice affects the estimated propensity score and outcome function. The directional conclusion (positive effect) is robust; the magnitude is not. We prioritize the RF estimate because it is closer to the experimental benchmark, but a rigorous analysis must present both and acknowledge this specification sensitivity openly rather than cherry-pick the favorable result.

External Validity

The NSW data is from 1975–1978. The U.S. labor market, training technologies, welfare policies, and workforce demographics have changed substantially. Our estimate tells us about the effect of NSW-style training on 1970s disadvantaged workers — extrapolating to modern programs such as the Workforce Innovation and Opportunity Act requires strong assumptions about transportability. For a contemporary policy recommendation, the analysis should ideally be re-estimated using modern data on comparable populations.

Return on Investment (ROI)

The NSW program cost approximately \$12,500 per participant in 1978 dollars. Our estimated annual earnings gain of \$1,541 implies a simple payback period of roughly 8 years if the effect persists. If the effect decays rapidly — as some Lalonde literature suggests — the ROI is weaker. A full ROI analysis would require estimating the persistence of the treatment effect over multiple post-program years, which our single post-program outcome variable (re78) cannot directly address. Our ROI claim should therefore be interpreted as conditional on sustained effects.

4. What I Cannot Rule Out

I cannot rule out that the positive DML estimate partly reflects unobserved positive selection — perhaps those who enrolled in NSW were more motivated to improve their economic situation, and this motivation would have driven earnings gains regardless of the program. The Lalonde literature, particularly Smith & Todd (2005), has shown that different comparison groups (CPS-1, CPS-3, PSID) yield materially different observational estimates, highlighting sensitivity to sample construction itself.

My estimate should be interpreted as the effect of training **conditional on observed covariates being sufficient to remove confounding**, rather than a definitive causal effect. The experimental benchmark provides reassurance, but the observational analysis on its own cannot claim the same internal validity as the RCT. A defensible policy recommendation must acknowledge this gap — not hide it.